

ServiceNow LDAP Integration

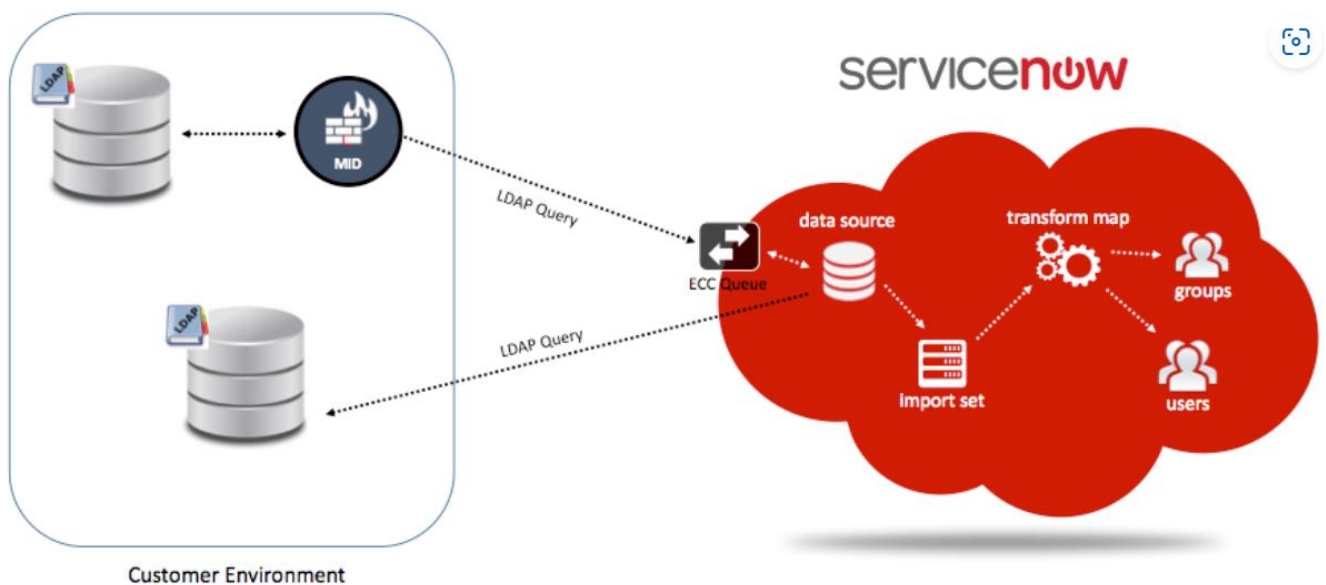
Topics Covered

- What is LDAP
- Why LDAP
- Basic LDAP Terms
- ServiceNow LDAP Integration with Test Server



What is LDAP?

- Lightweight Directory Access Protocol (LDAP).
- It is a protocol or a guide that helps applications to request for user information.
- LDAP in ServiceNow is a way to connect to external systems or Directories to manage users and groups.
- It helps ServiceNow authenticate users, import their information, and keep everything updated.





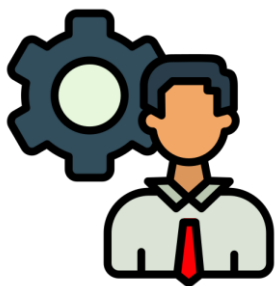
Why LDAP?

- Easy User Management.
- Centralized Authentication.
- Keeping data Synchronized with Directories.

Import users, groups from your existing directory or Active Directory making it easier to manage access and permissions.

Helps users to log in to ServiceNow using the same credentials they use for other systems, improving security and reducing password resets.

Ensure that user and group information are Synchronized with your directory regularly, without needing manual updates with help of Scheduled Data imports.



User Management



Centralized Auth



Data Synchronization

Basic LDAP Terms

- **Distinguished name (DN)**

This is a unique identifier of each entry or record in the Directory.

- **Relative distinguished name (RDN)**

The Starting search directory or RDN (Relative Distinguished Name) where you want Service-now to begin its search for users and/or groups.

- **Organizational Unit (OU)**

Helps organize the directory structure.

- **Common Name (CN)**

It is the name of the object (like a user).

- **User ID (UID)**

It is the unique identifier for a user or entry in the directory.

ServiceNow LDAP Integration with Test Server

Pre-requisite

- ServiceNow Instance
- Any Test Server with Credentials
Link for open test server [Online LDAP Test Server - Forum Systems](#)
- Follow the instructions.

Create LDAP Server

Navigator Bar > Create New Server

☐ Other

Server name
 ▶ More information

Server URL
 ▶ More information

Starting search directory
 ▶ More information

[Submit](#)

- Fill DN and Password according to the Test server Creds.

DN= *cn=read-only-admin,dc=example,dc=com*

Password= password

LDAP Server
Your wish

[o]

① Connected Successfully

Name

Active ☒

Application Global

Login distinguished name

Login password

Starting search directory

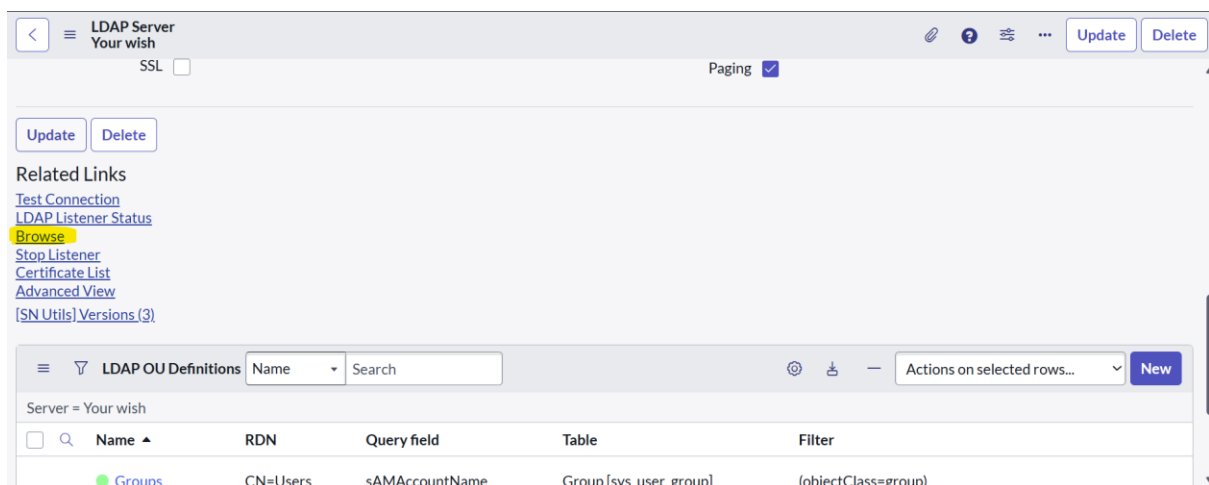
MID Server

LDAP Server URLs

URL	Order	Active	Operational Status
✗ ldap://ldap.forumsys.com	100	true	true

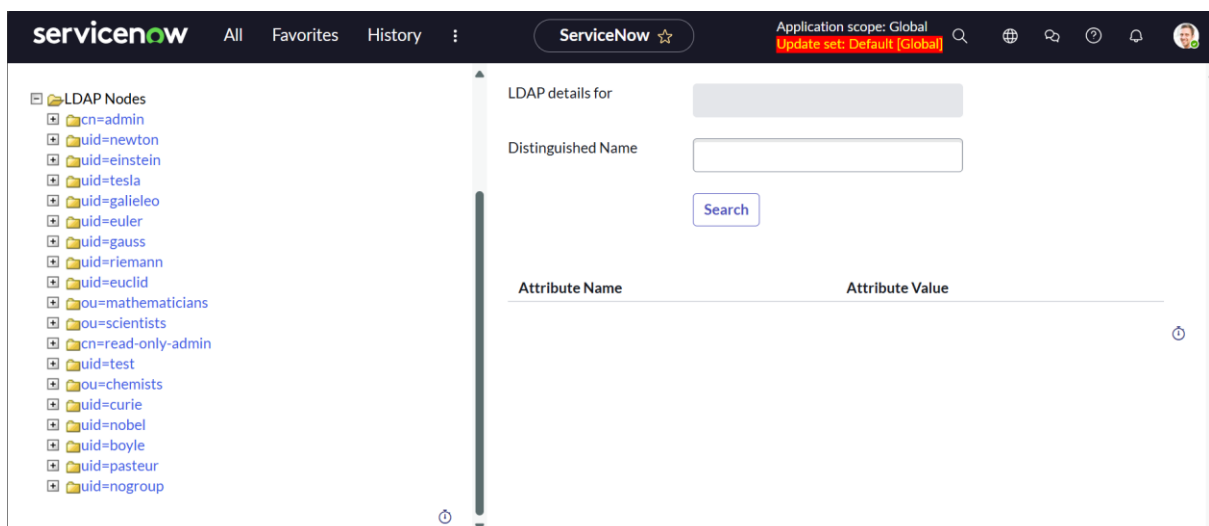
Successfully Connect to the Test Server

Now Click on Browse link in the Related Links



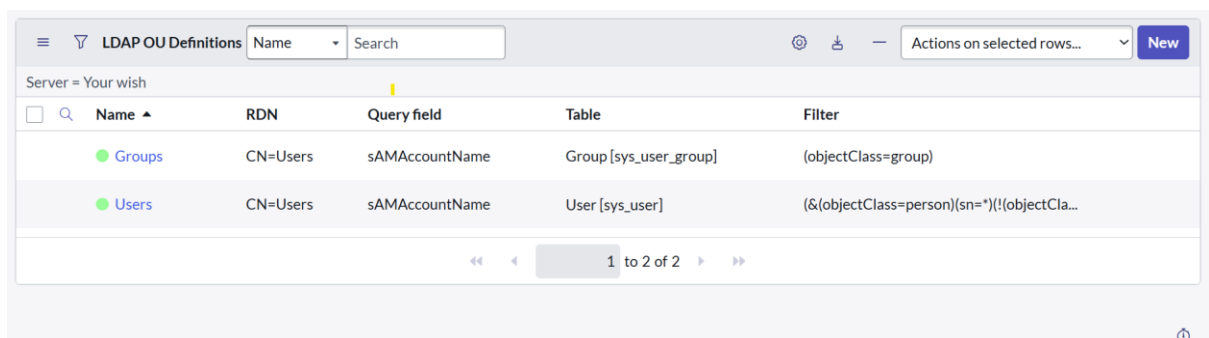
The screenshot shows the 'LDAP Server' interface for 'Your wish'. At the top, there are 'Update' and 'Delete' buttons. Below them is the 'Related Links' section, which includes links for 'Test Connection', 'LDAP Listener Status', 'Browse' (highlighted in yellow), 'Stop Listener', 'Certificate List', 'Advanced View', and '[SN Utils] Versions (3)'. Below the links is a table titled 'LDAP OU Definitions' with columns: Name, RDN, Query field, Table, and Filter. The table contains one entry for 'Groups' with RDN 'CN=Users', Query field 'sAMAccountName', Table 'Group [sys_user_group]', and Filter '(objectClass=group)'. There are also 'Update' and 'Delete' buttons for this entry.

Now we can See all the OU and UID's present in the Directory



The screenshot shows the 'ServiceNow' interface for 'LDAP Nodes'. On the left, there is a list of LDAP nodes including 'cn=admin', 'uid=newton', 'uid=einstein', 'uid=tesla', 'uid=galileo', 'uid=euler', 'uid=gauss', 'uid=riemann', 'uid=euclid', 'ou=mathematicians', 'ou=scientists', 'cn=read-only-admin', 'uid=test', 'ou=chemists', 'uid=curie', 'uid=nobel', 'uid=boyle', 'uid=pasteur', and 'uid=nogroup'. On the right, there is a search bar labeled 'LDAP details for' and a 'Search' button. Below the search bar is a table with columns 'Attribute Name' and 'Attribute Value'.

Go back to LDAP Server and open LDAP OU Definitions for Users



The screenshot shows the 'LDAP Server' interface for 'Your wish'. The 'LDAP OU Definitions' table now has two entries: 'Groups' and 'Users'. The 'Users' entry has RDN 'CN=Users', Query field 'sAMAccountName', Table 'User [sys_user]', and Filter '(&(objectClass=person)(sn=*))!(objectCla...'. There are also 'Update' and 'Delete' buttons for this entry.

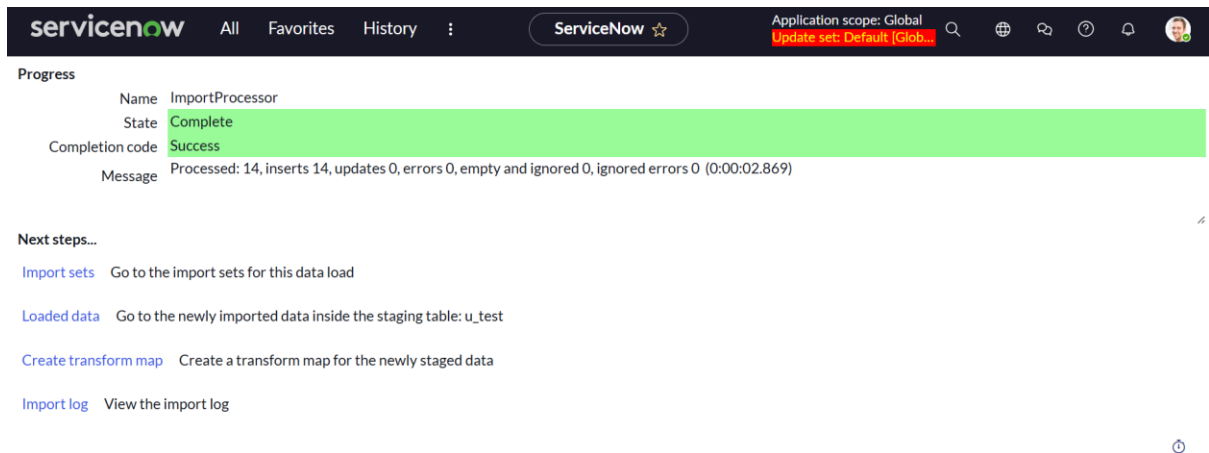
Clear the filter and (uid=*) to load all the users.

Then Click on the Data Source and load the records

The screenshot shows the ServiceNow interface for defining an LDAP OU. The top section is the 'LDAP OU Definition - Users' form. It includes fields for Name (Users), RDN, Query field (sAMAccountName), Filter ((uid=*)) and Application (Global). There are also fields for Active (checked), Server (Your wish), and Table (User [sys_user]). Below the form are 'Update' and 'Delete' buttons and a 'Related Links' section with links for 'Test connection', 'Browse', and '[SN Utils] Versions (3)'. The bottom section is a table titled 'Data Sources' with a search bar and a table header 'LDAP target = Users'. The table has two columns: 'Name' and 'Import set table name'. It contains one row with 'Your wish/Users' and 'u_test'. The table shows '1 to 1 of 1' records.

Now click on Load all records

The screenshot shows the ServiceNow interface for defining a Data Source. The top section is the 'Data Source - Your wish/Users' form. It includes fields for Name (Your wish/Users), Application (Global), Import set table label (test), Import set table name (u_test), Type (LDAP), and LDAP target (Users). There is a checkbox for 'Use Batch Import' which is unchecked. Below the form are 'Update' and 'Delete' buttons and a 'Related Links' section with links for 'Test Load 20 Records', 'Load All Records', and 'Run Point Scan'.



The screenshot shows the ServiceNow interface with a dark header. The main content area is titled 'Progress' and displays details for an 'ImportProcessor'. The 'State' is 'Complete' and the 'Completion code' is 'Success'. A message indicates 'Processed: 14, inserts 14, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:02.869)'. Below this, 'Next steps...' are listed: 'Import sets' (Go to the import sets for this data load), 'Loaded data' (Go to the newly imported data inside the staging table: u_test), 'Create transform map' (Create a transform map for the newly staged data), and 'Import log' (View the import log).

ServiceNow All Favorites History ServiceNow Application scope: Global Update set: Default [Glob...]

Progress

Name ImportProcessor

State Complete

Completion code Success

Message Processed: 14, inserts 14, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:02.869)

Next steps...

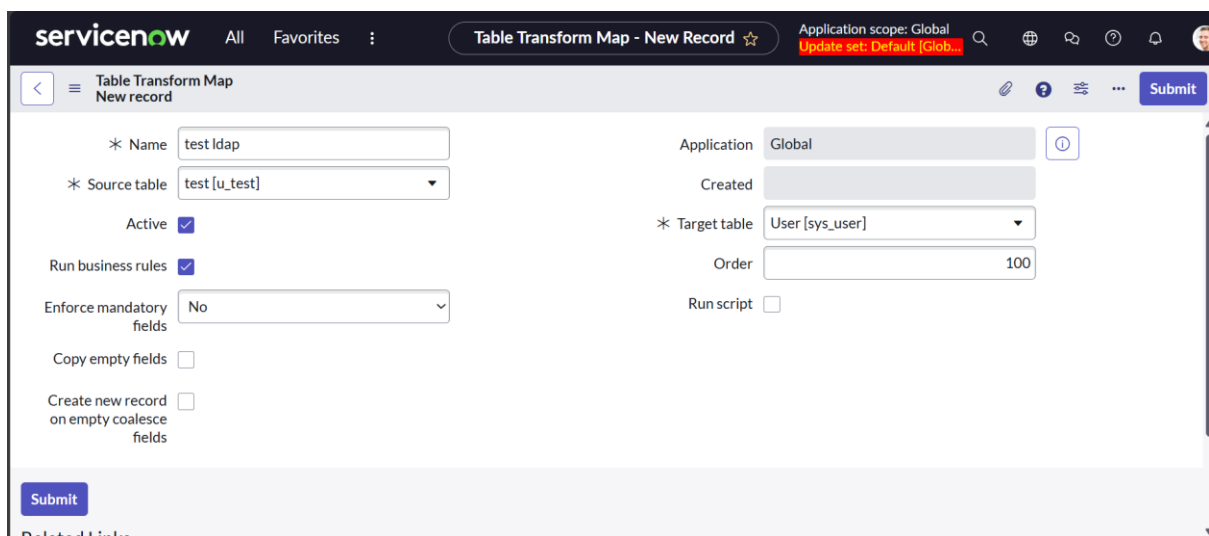
[Import sets](#) Go to the import sets for this data load

[Loaded data](#) Go to the newly imported data inside the staging table: u_test

[Create transform map](#) Create a transform map for the newly staged data

[Import log](#) View the import log

Click on create transform map create a staging table and set target table to sys_user table then map the field and run the transform maps



The screenshot shows the 'Table Transform Map - New Record' form in ServiceNow. The form has two columns. The left column contains fields for 'Name' (test ldap), 'Source table' (test [u_test]), 'Active' (checked), 'Run business rules' (checked), 'Enforce mandatory fields' (No), 'Copy empty fields' (unchecked), and 'Create new record on empty coalesce fields' (unchecked). The right column contains fields for 'Application' (Global), 'Created' (empty), 'Target table' (User [sys_user]), 'Order' (100), and 'Run script' (unchecked). A 'Submit' button is at the bottom left.

ServiceNow All Favorites History Table Transform Map - New Record Application scope: Global Update set: Default [Glob...]

Table Transform Map
New record

* Name test ldap

* Source table test [u_test]

Active ☒

Run business rules ☒

Enforce mandatory fields No

Copy empty fields ☐

Create new record on empty coalesce fields ☐

Application Global

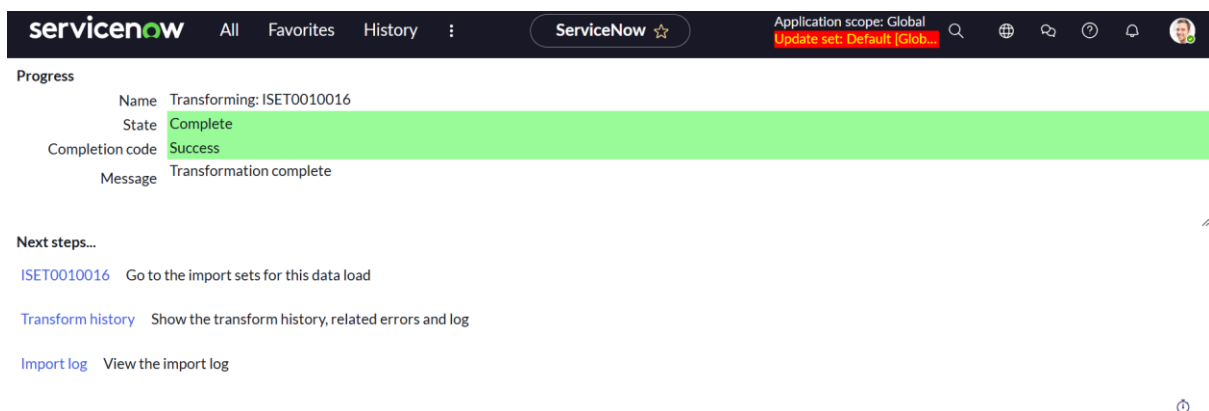
Created

* Target table User [sys_user]

Order 100

Run script ☐

Submit



The screenshot shows the ServiceNow interface with a dark header. The main content area is titled 'Progress' and displays details for a transformation job 'Transforming: ISET0010016'. The 'State' is 'Complete' and the 'Completion code' is 'Success'. The message indicates 'Transformation complete'. Below this, 'Next steps...' are listed: 'ISET0010016' (Go to the import sets for this data load), 'Transform history' (Show the transform history, related errors and log), and 'Import log' (View the import log).

ServiceNow All Favorites History ServiceNow Application scope: Global Update set: Default [Glob...]

Progress

Name Transforming: ISET0010016

State Complete

Completion code Success

Message Transformation complete

Next steps...

[ISET0010016](#) Go to the import sets for this data load

[Transform history](#) Show the transform history, related errors and log

[Import log](#) View the import log

Transform Histories

Started

Search

Actions on selected rows...

All > Set Number = ISET0010016

☐	Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skip
☐	2025-02-09 07:55:31	Complete	2025-02-09 07:55:32	1 Second	ISET0010016	test [u_test]	14	14	0	0	

Users

Created

Search

Actions on selected rows...

New

All

User ID	Name	Email	Active	Created	Updated	Source
(empty)	Louis Pasteur		true	2025-02-09 08:01:36	2025-02-09 08:01:36	ldap:uid=pasteur,dc=example,dc=com
(empty)	Isaac Newton		true	2025-02-09 08:01:35	2025-02-09 08:01:35	ldap:uid=newton,dc=example,dc=com
(empty)	Alfred Nobel		true	2025-02-09 08:01:35	2025-02-09 08:01:35	ldap:uid=nobel,dc=example,dc=com
(empty)	Bernhard Riemann		true	2025-02-09 08:01:35	2025-02-09 08:01:35	ldap:uid=riemann,dc=example,dc=com
(empty)	Nikola Tesla		true	2025-02-09 08:01:35	2025-02-09 08:01:35	ldap:uid=tesla,dc=example,dc=com
(empty)	Albert Einstein		true	2025-02-09 08:01:35	2025-02-09 08:01:35	ldap:uid=einstein,dc=example,dc=com
(empty)	Marie Curie		true	2025-02-09 08:01:35	2025-02-09 08:01:35	ldap:uid=curie,dc=example,dc=com

Follow the same for groups
